

ACRME 4-Region DR Model — What-If Analysis Comparison

Document Purpose: Compare the baseline v2.4 region distribution models against a proposed 4-region upgrade for US and EU geographies.

Workbook: `acrme_region_selection_whatif_4region.xlsx` (what-if variant; does NOT modify baseline requirements)

Model Comparison Matrix

Geography	Baseline v2.4 Model	4-Region What-If	Regions Required	Impact
US	3-region	4-region	4 Standard regions	✓ All 4 US Standard regions compete
EU	2-region (CVAL/DR co-located)	4-region	4 Standard regions	✓ CVAL and DR no longer co-locate
Australia	2-region (CVAL/DR co-located)	2-region (unchanged)	2 Standard regions	No change
Asia Pacific	2-region (CVAL/DR co-located)	2-region (unchanged)	2 Standard regions	No change
Middle East	cross-geo DR (Prod+CVAL local; DR in EU)	cross-geo (unchanged)	1 local + 1 EU	No change

Key Changes — 4-Region Model

1. PLC-010a Co-Location Rule — DEACTIVATED for US and EU

Baseline (2-region model):

PLC-010a: In 2-region geographies (EU, Australia, Asia Pacific), CVAL and DR MUST co-locate in the same region.

4-Region model:

- **US and EU:** PLC-010a **NO LONGER APPLIES**

- Prod, CVAL, and DR can each occupy **separate, distinct regions**
- Improves **blast-radius isolation** — a CVAL region failure no longer impacts DR readiness

Australia, Asia Pacific: PLC-010a still applies (remain 2-region)

2. HC-6 and HC-7 Hard Constraint Gates — DISABLED for US/EU

HC Gate	Baseline (2-region)	4-Region Model (US/EU)
HC-3 Prod	<code>Prod_Q_Headroom >= Req_vCPU</code> <code>AND Prod_Q_Headroom - Req_vCPU >= MinBuffer</code>	✅ Same (still applies)
HC-3 NonProd	<code>NP_Eff_Free >= Req_vCPU AND NP_Q_Headroom - Req_vCPU >= MinBuffer</code>	✅ Same (still applies)
HC-3 DR	<code>DR_Free >= DR_bootstrap_qty</code>	✅ Same (still applies)
HC-6 DR Floor	<code>DR_Free + CVAL_Eff_Free >= DR_bootstrap_qty</code>	⚠️ N/A (4-region) — auto-satisfied
HC-7 DR Integrity	<code>CVAL_Q_Used + Req_vCPU <= CVAL_Q_Limit - DR_bootstrap_qty</code>	⚠️ N/A (4-region) — auto-satisfied

Implementation:

- **HC-6/HC-7 formulas** check geography first:

excel

```
=IF(OR(Geography="US",Geography="EU"), "N/A (4-region)", <normal check>)
```

- Eligibility logic treats "N/A (4-region)" as an automatic **PASS**
- **Result:** US and EU regions are **no longer blocked** by CVAL/DR co-location constraints

3. Region Catalogue — All Standard Regions Now Compete

Geography	Baseline Restricted Regions	4-Region Model Status
US	East US 2 (exception required)	✅ Promoted to Standard (competes for placement)
EU	North Europe, West Europe (exception required)	✅ Promoted to Standard (competes for placement)
Middle East	Saudi Arabia East, UAE North (production-only)	⚠️ Remain Restricted (unchanged)

Impact:

- **US**: 4 candidate regions (West US 3, Central US, Canada Central, East US 2)
- **EU**: 4 candidate regions (Switzerland North, Sweden Central, North Europe, West Europe)

4. Region Selection Sequencing — 3-Pass vs 2-Pass

Baseline (2-region EU):

1. Select Prod region (weighted scoring on 2 Standard regions)
2. Select CVAL/DR region (the other Standard region — they co-locate)

4-Region model (EU):

1. Select Prod region (weighted scoring on all 4 Standard regions)
2. Select CVAL region (weighted scoring on remaining 3 regions, excluding Prod)
3. Select DR region (weighted scoring on remaining 2 regions, excluding Prod and CVAL)

Impact: More **fine-grained placement** — each environment gets its own weighted-optimal region

5. Capacity Accounting — Simplified (US/EU)

Baseline (2-region co-located model):

- CVAL and DR share the same **Reserved** pool
- Complex accounting: HC-6 (combined DR+CVAL floor), HC-7 (CVAL allocation protects DR headroom)
- Risk: CVAL allocation can consume DR reserve

4-Region model (US/EU):

- Each environment (Prod / CVAL / DR) has its **own Reserved pool**
- Independent capacity accounting — no cross-environment interference
- Simpler HC checks: each region evaluated purely on its own metrics

6. DR Sizing Formula — No Change

The **MAX-NOT-SUM** sizing rule (A.6 / DR-017) remains unchanged:

$$\text{DR_req} = \text{MAX over all source_regions of } (\text{customers_in_source} \times \text{vCPU_per_customer})$$

- This is **model-independent**
- **Multi-source DR hosting** (DR-016) benefits from 4-region model (more DR destination candidates)

7. Placement Scoring Engine — γ Component Adjustment

Component	Baseline (2-region EU)	4-Region Model (EU)
α (Efficient Free)	Same formula	Same formula
β (Quota Headroom)	Same formula	Same formula
γ (Distribution Fairness)	$1 - (\text{customers_in_region} / \text{total_customers})$ normalized over 2 regions	$1 - (\text{customers_in_region} / \text{total_customers})$ normalized over 4 regions
δ (DR Coverage)	CVAL region scores for DR too	Separate DR scoring
ϵ (Zone Diversity)	Same formula	Same formula

Impact: γ component normalizes over **4 regions** instead of 2, improving distribution fairness

What-If Workbook Usage

1. **Open** `acrm_region_selection_whatif_4region.xlsx`
2. **Navigate to Setup sheet:**
 - Select **Geography** (US or EU to see 4-region impact)
 - Enter **SKU, VM count, Customer ID**
3. **Review HC_Gate sheet:**
 - US and EU regions show **“N/A (4-region)”** for HC-6 and HC-7
 - All US/EU regions automatically eligible for NonProd and DR (if HC-3 passes)
4. **Check Scoring sheets:**
 - More regions compete (4 candidates instead of 2-3)
 - Rank column shows placement order
5. **Review Result sheet:**
 - Prod, CVAL, DR placements (may occupy **3 distinct regions** in 4-region model)
 - Readiness states
 - Ranked Candidates table

Side-by-Side Comparison: US Geography Example

Baseline (3-region model)

- **Prod:** West US 3 (Rank 1, PS = 0.82)
- **CVAL:** Central US (Rank 2, PS = 0.76)
- **DR:** Canada Central (Rank 3, PS = 0.71)
- **East US 2:** Excluded (Restricted — requires exception)

4-Region Model

- **Prod:** West US 3 (Rank 1, PS = 0.82)

- **CVAL**: Central US (Rank 2, PS = 0.76)
- **DR**: Canada Central (Rank 3, PS = 0.71)
- **East US 2**: Now **Rank 4** candidate (PS = 0.68) — competes for placement

Impact: East US 2 now participates in weighted scoring; if capacity shifts (e.g., West US 3 fills up), East US 2 automatically becomes a viable alternative.

Side-by-Side Comparison: EU Geography Example

Baseline (2-region model with PLC-010a co-location)

- **Prod**: Switzerland North (Rank 1, PS = 0.79)
- **CVAL + DR**: Sweden Central (Rank 2, PS = 0.74) — **co-located in the same region**
- **North Europe, West Europe**: Excluded (Restricted)

Constraints:

- HC-6: $DR_Free + CVAL_Eff_Free \geq DR_bootstrap_qty$ (joint check)
- HC-7: $CVAL_Q_Used + Req_vCPU \leq CVAL_Q_Limit - DR_bootstrap_qty$ (protects DR floor)

4-Region Model

- **Prod**: Switzerland North (Rank 1, PS = 0.79)
- **CVAL**: Sweden Central (Rank 2, PS = 0.74)
- **DR**: North Europe (Rank 3, PS = 0.69) — **separate region**
- **West Europe**: Now Rank 4 candidate (PS = 0.65)

Constraints:

- HC-6: **N/A (4-region)** — auto-satisfied
- HC-7: **N/A (4-region)** — auto-satisfied
- CVAL and DR evaluated independently

Impact:

1. **Blast-radius improvement**: DR no longer impacted by CVAL region failure
 2. **Capacity independence**: CVAL allocation cannot consume DR reserve
 3. **More placement options**: 4 regions compete instead of 2
-

Migration Path (If Adopted)

If the 4-region model is adopted as the new baseline:

Phase 1: US (3-region → 4-region)

- ✓ **Low risk** — US already has 3 separated regions
- Promote **East US 2** from Restricted → Standard
- **Option A**: Dual-DR model (Prod / CVAL / DR-Primary / DR-Secondary)
- **Option B**: 4-way split (Prod / CVAL / DR-1 / DR-2)

Phase 2: EU (2-region → 4-region)

- ⚠ **Higher impact** — requires breaking CVAL/DR co-location
- Promote **North Europe + West Europe** from Restricted → Standard

- Update **REG-003** (distribution model) to add “4-region” as a third model
- Amend **PLC-010** to carve out US/EU from PLC-010a (co-location rule)
- Remove **HC-6/HC-7** co-location checks for 4-region geographies
- **Migrate existing EU customers:** re-run placement seed (PLC-003) to split CVAL and DR into separate regions

Requirements Impact Summary (If Adopted)

Requirement	Current State	4-Region Model Change
REG-003	Three models: 3-region (US), 2-region (EU/AU/AP), cross-geo (ME)	Add 4-region model for US and EU
PLC-010a	CVAL/DR co-location mandatory in 2-region geographies	Carve out US/EU (no longer applies to 4-region)
HC-6	DR Floor check for 2-region co-located model	Disabled for 4-region geographies
HC-7	DR Integrity check for 2-region co-located model	Disabled for 4-region geographies
ENV-003	Co-location-in-region vs capacity-sharing clarified	No change (still applies to Australia, Asia Pacific)

Known Limitations (What-If Model)

1. **No real production data** — workbook uses SYNTHETIC distributed capacity
2. **East US 2, North Europe, West Europe** promoted to Standard for this what-if only (still Restricted in baseline v2.4)
3. **Migration plan not included** — existing EU customers would need placement re-seeding
4. **No cost model** — does not quantify 4-region DR reserve cost vs 2-region co-located model

Next Steps

1. **Run the what-if workbook** with representative customer scenarios (US, EU)
2. **Compare placement results:** baseline vs 4-region model
3. **Evaluate blast-radius benefits** (CVAL/DR separation in EU)
4. **Assess cost impact** (4 DR regions vs 2 co-located regions)
5. **Decision:** formal REG-003 amendment or remain with baseline v2.4 models

Document Status: What-if exploration only — does NOT modify baseline requirements

Baseline Reference: Requirements Baseline v2.4, REG-003, PLC-010a, HC-6, HC-7

Workbook: `acrme_region_selection_whatif_4region.xlsx`

Builder Script: `build_acrme_whatif_4region.py`